

CI/CD Fundamentals

Building, Testing, and Deploying with Confidence

What is CI/CD?

Continuous Integration (CI)

- Automatically build and test code changes
- Merge code frequently to main branch
- Detect integration issues early

Continuous Delivery (CD)

- Automatically preparing code for release
- Code is always deployment-ready
- Manual approval for production

Why Use CI/CD?

- **Fast Feedback** - Know immediately if code breaks
- **Reduced Risk** - Smaller changes easier to debug
- **Consistent Quality** - Automated tests every time
- **Faster Releases** - Eliminate manual steps
- **Better Collaboration** - Catch issues early

Common CI/CD Providers

Cloud-Based

- GitHub Actions
- GitLab CI/CD
- CircleCI
- Travis CI
- Azure DevOps
- AWS CodePipeline

Self-Hosted Options

- Jenkins
- TeamCity
- Bamboo
- Drone CI
- Buildkite

Most Popular for New Projects: GitHub Actions (integrated) or GitLab CI/CD

What is GitHub Actions?

GitHub's built-in CI/CD platform that automates workflows directly in your repository.

Key Features:

- Triggered by GitHub events (push, PR, schedule)
- Runs on GitHub-hosted or self-hosted runners
- YAML-based configuration
- Free for public repos

Location: `.github/workflows/` directory

Anatomy of a GitHub Actions Workflow

```
name: CI Pipeline                                # Workflow name

on:                                              # Trigger events
  push:
    branches: [main]
  pull_request:

jobs:                                           # One or more jobs
  build:
    runs-on: ubuntu-latest                     # Runner environment

    steps:                                     # Sequential steps
      - uses: actions/checkout@v4             # Pre-built action

      - name: Install dependencies             # Custom step
        run: npm install

      - name: Run tests
        run: npm test
```

GitHub Actions Triggers

Common Triggers:

```
on:  
  push:  
    branches: [main, develop]  
  
  pull_request:  
    types: [opened, synchronize]  
  
  schedule:  
    - cron: '0 0 * * *'  
  
  workflow_dispatch: # Manual
```

When to Use:

- `push` - Deploy to environments
- `pull_request` - Run tests, linting
- `schedule` - Nightly builds, cleanups
- `workflow_dispatch` - Manual deployments
- `release` - Publish packages
- `issue_comment` - Bot interactions

GitHub Actions Marketplace

Reusable actions created by GitHub and the community.

Popular Actions:

- `actions/checkout@v4` - Clone repo
- `actions/setup-node@v4` - Setup Node.js
- `actions/upload-artifact@v4` - Save build outputs
- `codecov/codecov-action` - Upload coverage

Using Actions:

```
steps:  
  # Use specific version  
  - uses: actions/checkout@v4  
  
  # With parameters  
  - uses: actions/setup-node@v4  
    with:  
      node-version: '20'  
  
  # Multiple versions  
  - uses: actions/cache@v4  
    with:  
      path: ~/.npm  
      key: ${ runner.os }-npm
```


Managing Secrets in Pipelines

Why Secrets Management Matters:

- API keys, passwords shouldn't be in code
- Different secrets per environment
- Secure access to external services
- Compliance and security requirements

Best Practices:

- Never commit secrets to version control
- Use secret stores
- Rotate secrets regularly
- Limit access with least privilege
- Audit secret usage

Managing Secrets in GitHub

GitHub Secrets Types:

- **Repository** - One repo
- **Environment** - Per environment
- **Organization** - Shared across repos

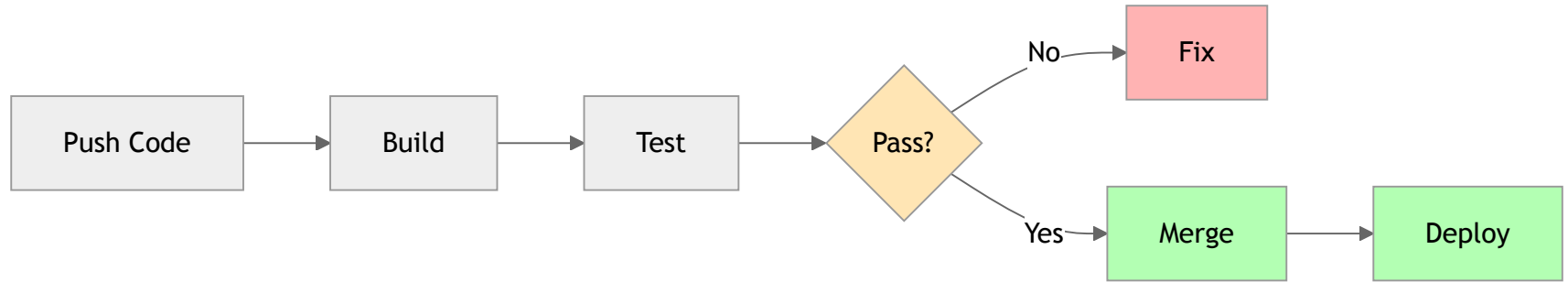
Using Secrets:

```
steps:  
  - name: Deploy  
    env:  
      API_KEY: ${ secrets.API_KEY }  
      DATABASE_URL: ${ secrets.DATABASE_URL }  
    run: npm run deploy
```

Set: Settings → Secrets and variables → Actions

Note: Secrets are masked in logs!

Typical CI/CD Pipeline



CI/CD Best Practices

Speed & Efficiency:

- Keep builds fast (< 10 minutes ideal)
- Run tests in parallel
- Fail fast - critical steps first

Reliability:

- Keep main branch always green
- Fix broken builds immediately
- Use branch protection rules

Security:

- Scan for vulnerabilities
- Never log sensitive information
- Use secrets management

Common Challenges

Flaky Tests

- Tests that randomly fail
- Solution: Retry logic, better isolation

Slow Builds

- Long pipeline execution
- Solution: Caching, parallelization

Secret Management

- Handling credentials safely
- Solution: Built-in secret stores

Environment Differences

- "Works on my machine"
- Solution: Docker, consistent environments

Debugging Failures

- Hard to reproduce locally
- Solution: Good logging, artifacts

Dependency Issues

- Version conflicts
- Solution: Lock files, containers

Reading Build Logs

When a build fails:

1. **Check the failing step** - Which command failed?
2. **Read error messages** - Often clear about the issue
3. **Check previous successful builds** - What changed?
4. **Review recent commits** - New code causing issues?

Branch Protection Rules

Enforce quality through GitHub settings:

Typical Configuration:

-  Require pull request before merging
-  Require status checks to pass (CI, tests, linting)
-  Require branches to be up to date
-  Require code review approval
-  Dismiss stale reviews on new commits

Result: Can't merge broken code into main!

Quality Gates

Define when builds should fail:

Code Quality:

- Linting errors
- Code coverage below threshold
- TypeScript errors
- Code complexity too high

Testing:

- Any test failure
- Coverage drops
- E2E test failures

Security:

- Known vulnerabilities
- Security scan failures

Challenge: Your First CI Pipeline

Time to put knowledge into practice!

Exercise Task

Goal: Create a GitHub Actions workflow that validates code quality on every pull request.

Scenario: Your team wants to ensure all code changes are:

- Properly formatted
- Free of linting errors
- Passing all tests
- Maintains 80% code coverage for tests

Your Task: Build a CI pipeline that automatically checks these before allowing merges.

Exercise Requirements

Workflow Triggers:

- Must run on pull request events
- Should run on pushes to `main` branches

Success Criteria: All steps must pass for the workflow to succeed

Exercise Hints

File Location:

- Create workflow file in `.github/workflows/` directory
- Name it something like (e.g., `ci.yml` or `pr-checks.yml`)

Useful Actions:

- `actions/checkout@v4` - Clone repo
- `actions/setup-node@v4` - Setup Node.js

Common Commands:

- `npm install` or `npm ci` - Install dependencies

Key Takeaways

- **CI/CD automates** build, test, and deployment
- **Fast feedback** helps catch issues early
- **GitHub Actions** provides powerful CI/CD in `.github/workflows/`
- **Secrets management** keeps credentials secure
- **Branch protection + CI** prevents broken code merging
- **Caching** speeds up pipelines
- **Start simple** - add complexity as needed

Questions?

Time to set up your first GitHub Actions workflow!